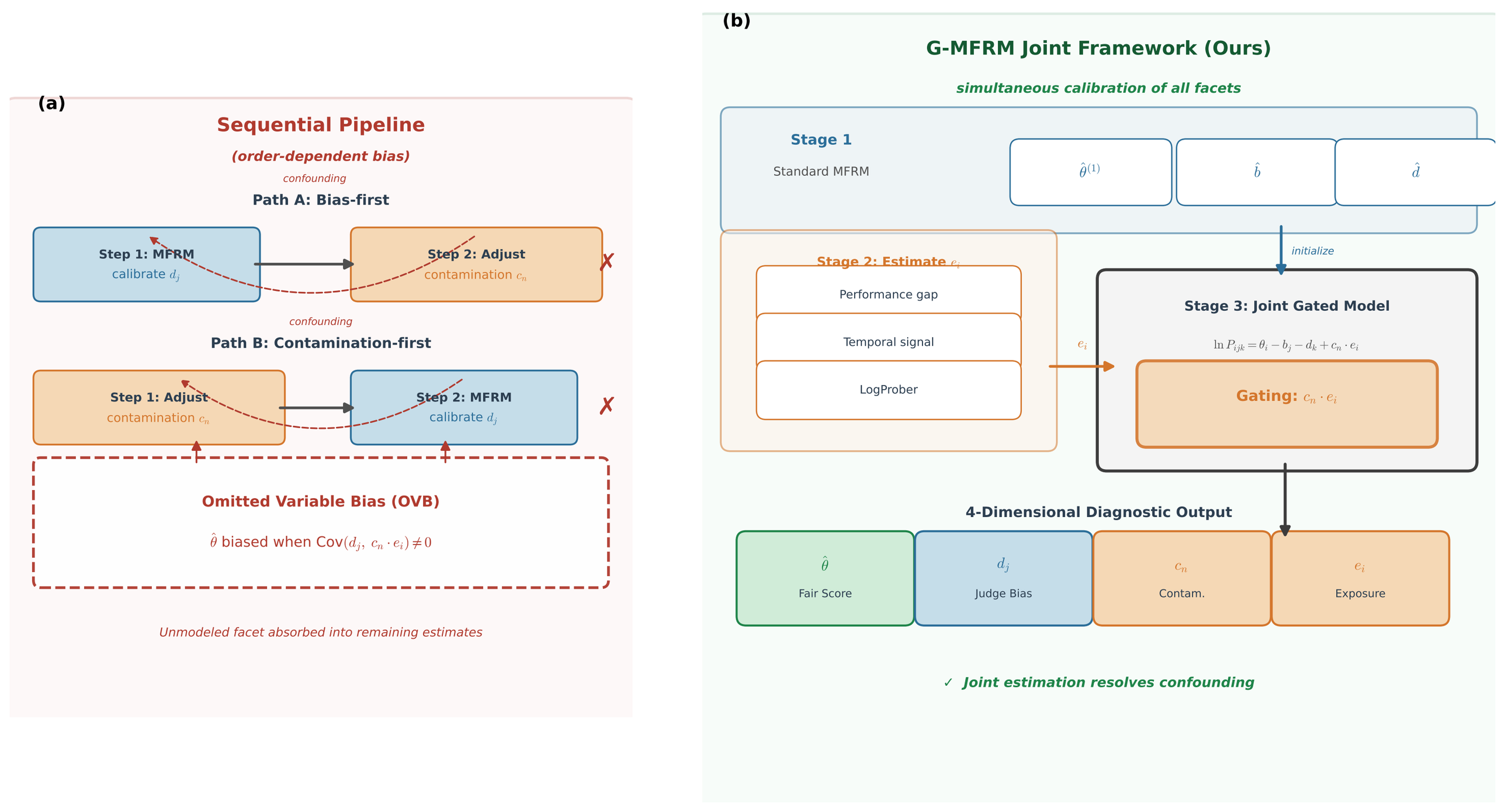

(a)
Sequential Pipeline
(order-dependent bias)
confounding
Path A: Bias-first
Step 1: MFRM
calibrate d_j
Step 2: Adjust
contamination c_n
confounding
Path B: Contamination-first
Step 1: Adjust
contamination c_n
Step 2: MFRM
calibrate d_j
Omitted Variable Bias (OVB)
$\hat{\theta}$ biased when $\mathrm{Cov}(d_j,\ c_n \cdot e_i) \neq 0$
Unmodeled facet absorbed into remaining estimates
(b)
G-MFRM Joint Framework (Ours)
simultaneous calibration of all facets
Stage 1
Standard MFRM
$\hat{\theta}^{(1)}$
$\hat{b}$
$\hat{d}$
initialize
Stage 2: Estimate e_i
Performance gap
Temporal signal
LogProber
e_i
Stage 3: Joint Gated Model
$\ln P_{ijk} = \theta_i - b_j - d_k + c_n \cdot e_i$
Gating: $c_n \cdot e_i$
4-Dimensional Diagnostic Output
$\hat{\theta}$
Fair Score
d_j
Judge Bias
c_n
Contam.
e_i
Exposure
✓ Joint estimation resolves confounding